

MOST HAPPENING, FACT-FILLED CURRENT AFFAIRS FROM AROUND THE WORLD

3D printed robot kit

A 3D printed robot kit, Mobot-A, has been launched by Barobo Inc., which offers students the opportunity to learn 21st century skills by building a robot from the ground up.

"As 3D printers become more and more common in the classrooms, there is a need for engaging projects and curriculum to tie this powerful tool into science, technology, engineering and math (STEM) subjects," said Graham Ryland, president and co-founder of Barobo Inc.

The Mobot-A kit includes the internal electronics, motors, and fasteners, and once assembled, it can be attached to other robots and accessories to form new machines. Students can design their own accessories to attach to the robot and print on a 3D printer.

"We're breaking from traditional business models and relying on our users to not just assemble the robot but play an active role in manufacturing the plastic parts," added Ryland.

LED clip that dries clothes

Panasonic has changed the humble image of the cloth clips by unveiling its BF-AF20P LED clip, which can bring a revolution in the world of dry cleaning. The clip can not only be used as a handy torch to illuminate surrounding areas, but the attached LED is also designed to actively speed up the drying process when users peg up their clothes to dry.

The Japan-based company has taken into account all possible issues with the new gadget, cleverly making it waterproof, which is a necessary attribute when aiming to accomplish the task of helping users keep wet clothes on a clothesline.

The device comes in an array of different colours to suit buyers' preference including yellow, pink and blue. The LED light runs on two CR2032 button batteries, and measures 35mm



Barobo Mobot-A robot kit allows students to build 3D printed robots

× 34mm × 50mm, tipping the scales at approximately 19 gm, with batteries included.

Indian scientists develop world's smallest antenna

Scientists at the Birla Institute of Technology, Mesra, claim to have developed the world's smallest antenna to be used in homes, military and other domains.

The antenna is a super compact ultra-wideband (UWB) planar antenna. The UWB technology brings mobility of wireless communications with high data rates, designed for short range, and wireless personal area networks (WPANs) freeing users from the constraints of having to use a wired connection.

The scientists said that the antenna is, "the world's smallest at 14 mm × 11 mm, with much more than a 10:1 bandwidth."

The antenna is cheap to produce, with the goal now being to reduce the size of the device further so that it can be pasted on any curved surface when it eventually becomes available for general purchase.

A battery with power up to 25 kW

Researchers all over the world are working towards finding alternate sources of generating electricity such as solar and wind energy. It is important to have powerful electric energy storage devices to level out corresponding irregularities in the power supply.

Scientists at the Fraunhofer Institute for Environmental, Safety and Energy Technology UMSICHT in Oberhausen, Germany, have recently made an important breakthrough by developing a redox flow battery that reaches stack power up to 25 kW, with a cell size of 0.5 square meters. This is eight times larger than the previous A4-sized systems.

Redox flow batteries offer an effective way to balance out fluctuations in the supply of renewable energy and thus guarantee its constant availability. The batteries store electrical energy in chemical compounds, the liquid electrolytes. The electrolytes are charged and discharged in small reaction cham-